Homework — 1.3.3 Networks — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [2] Any two for 1 mark each: a LAN covers a small geographical area / single site, a WAN covers a large area (cities/countries); a LAN's hardware is owned by the organisation, a WAN uses third-party/leased infrastructure; LANs are typically faster. (The internet is the largest WAN.)

2. (a) [1] Any one: a single cable fault only affects the one device on that cable (others keep working) / easy to add or remove devices / good performance as data goes via the central switch (no collisions with a switch).

(b) [1] Any one: the central switch/hub is a single point of failure — if it fails the whole network goes down / requires more cabling than some other layouts / cost of the central device.

3. [3] Up to 3 from: the browser sends the typed domain name to a DNS server **[1]**; the DNS looks up / translates the domain name into the server's IP address (querying a hierarchy of DNS servers if needed, results may be cached) **[1]**; the IP address is returned to the browser **[1]**; the browser then uses that IP address to send its request to the correct web server **[1]** (max 3).

4. [3] Two advantages (1 mark each): centralised security / user accounts / backups; easier central management and software updates; central data storage and control; scales well to many users. One disadvantage (1 mark): the server is a single point of failure / more expensive (server hardware and administration needed) / can become a bottleneck. (Must give two advantages and one disadvantage.)

5. [3] 1 mark: POP3 downloads messages to the local device and (typically) removes them from the server. 1 mark: IMAP keeps the messages on the server and synchronises across devices (so the same mailbox is seen from anywhere). 1 mark: SMTP is used to send mail.

6. [1] A firewall controls/monitors incoming and outgoing network traffic against a set of rules (packet filtering), blocking traffic that is not permitted to protect the network from unauthorised access.

Part B (6 marks)

7. [2] 1 mark: every foreign key must reference a valid existing primary key in the related table (or be null). 1 mark: this prevents orphaned records / ensures links between tables remain valid (e.g. you cannot add a record that refers to one that does not exist, or delete a record other records still depend on).

8. [2] Any two for 1 mark each: one-way / not reversible; deterministic (same input → same hash); few/low collisions; fast to compute; produces a fixed-length output.

9. [2] 1 mark + 1 mark: the linker combines the compiled program with the separately compiled library modules / object files it uses **[1]** and resolves the references/addresses between them so they work together as a single executable **[1]**. (Accept: combining multiple object files; including external library code.)

Total: 14 + 6 = 20 marks